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Odd Semester End Term Examination, December 2024

Faculty of Engineering, SCSE/SEEC

Department of CSE/AIIML/ECE/ELC/EEE/EVL

B. Tech.

Course Code: MAS2001

Course: Statistical Method and Probability Theory

Semester: III

Time: 03 hrs.

Max. Marks: 80

Instructions: All questions are compulsory.

Missing data, if any, may be assumed suitably.

Calculator is allowed.

SECTION A

- A1 If $n=50$ and $p=0.4$ then mean of binomial distribution is ... MP [2]
- A2 If $K=3$ in Chebyshev's inequality, what percentage of values lie within 3 standard deviations of the mean? [2]
- A3 If the expected value of an estimator is equal to its parametric function. It is said to be... Sufficient [2]
- A4 Testing $H_0: \mu = 1000$, against $H_1: \mu > 1000$ leads to [2]
a) One sided right-tailed test
b) One sided left-tailed test
c) Two tailed-test
d) None of these
- A5 If the variance of a population is 400 and the sample size is 100 then by the central limit theorem $\bar{X}$ follows the normal distribution with standard deviation $\frac{20}{\sqrt{100}}$ [2]

SECTION B

- B1 If the Chebyshev's inequality for the random variable X is given by [5]
 $P(-2 < X < 8) \geq \frac{21}{25}$, find $E(X)$ and $Var(X)$.
- B2 A continuous random variable X is distributed over the interval (0, 1) with probability density function (PDF) $f(x) = ax^2 + bx$, where a, b are constants. If the mean of X is 0.5, find the value of a and b . [5]
- B3 The probability that a bomb dropped from a plane will strike the target is $\frac{1}{5}$. If six bombs are dropped, find the probability that at least two will strike the target and exactly two will strike the target. [5]
- B4 If the sum of the mean and variance of a binomial distribution of 5 trials is $\frac{9}{5}$, find the probability mass function of the binomial distribution. [5]
- B5 A random blood sample for test of fasting sugar for 10 boys gave the following data in mg/dl. [5]
70, 120, 110, 101, 88, 83, 95, 107, 100, 98
Does this support the assumption of population mean of 100 mg/dl? Test at 5% level of significance. Also find the 95% reasonable range in which most of the mean fasting sugar test of the 10 boys lie. (Tabulated value 2.262) confidence

X Two batches each of 12 animals are taken to test the effectiveness of an inoculation. One batch was inoculated and the other batch was not inoculated. The frequencies of the dead and surviving animals are given below in both cases. Can the inoculation be regarded as effective against the disease? (Tabulated value 3.841)

	Dead	Survived
Inoculated	2	10
Not inoculated	8	4

SECTION-C

- Q1 a. Given the frequency function, $f(x, \theta) = \frac{1}{\theta}$, $0 \leq x \leq \theta$ and $f(x, \theta) = 0$, otherwise [5+5]
and that you are testing the null hypothesis $H_0: \theta = 1$ against $H_1: \theta = 2$, by means of a single observed value x . What would be the sizes of the Type I and type II errors, if you choose the interval (i) $0.5 \leq x$, (ii) $1 \leq x \leq 1.5$ as the critical regions?
- b. Find the maximum likelihood estimate for the parameter λ of a Poisson distribution on the basis of a sample of size n .
- C2 a. A group of 7 week old chicken reared on a high protein diet weight 12, 15, 11, 16, 14, 14, and 16 ounces. A second group of five chickens similarly treated, except that they receive a low protein diet, weight 8, 10, 14, 10 and 13 ounces. Test at 5% level of significance whether there is evidence that additional protein has increased the weight of chicken. (tabulated value= 2.33). [5+5]
- b. Let X_1, X_2, X_3 and X_4 be independent random variable such that $E(X_i) = \mu$ and $V(X_i) = \sigma^2, i = 1, 2, 3, 4$.
If $Y = \frac{X_1+X_2+X_3+X_4}{4}$, $Z = \frac{X_1+X_2+X_3+X_4}{5}$ and $T = \frac{X_1+2X_2+X_3-X_4}{4}$
Examine whether Y, Z and T are unbiased estimator of μ ? Which one is the best estimator?

SECTION-D

- D1 a. A random number X is distributed at random between the values 0 and 1 so that its probability density function is $f(x) = kx^2(1 - x^3)$, where k is a constant. Find the value of k . Using this value of k , find its mean and variance. [5+5]
- b. In a certain examination 2000 students appeared. The average marks obtained were 50% and standard deviation was 5%. How Many students do you expect to obtain more than 60 marks, assuming the marks to be distributed normally? ($\phi(2) = 0.4772$)

- D2 Three different kinds of food are tested on three groups of rats for 5 weeks. The objective is to check the difference in mean weight (in grams) of the rats per week. Test the hypothesis at 0.05 significance level to the following data: (Tabulated value 3.68) [10]

ANOVA

Food I	Food II	Food III
8	4	11
12	5	8
9	4	7
8	6	13
6	9	7
11	7	9
